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Editor's Introduction

Christophe Granet continues his series on designing dual-reflector antennas with this column. In this paper, he extends his June, 2002, contribution to add possible ways to specify the dual-offset Cassegrain and Gregorian reflectors. When you consult his June, 2002, article for the geometry, you will have 17 ways to specify these reflectors. He also includes the geometry of the feed horn in these designs. His earlier articles were published in this column: April, 1998: "Designing Axially Symmetric Cassegrain and Gregorian Dual-Reflector Antennas;" June, 1998: "Minimum Blockage in Dual-Reflector Antennas While Taking into Account the Phase-Center of the Feed;" December, 1999: "Displaced-Axis Dual Reflector Antennas" (four types); and December, 2001: "Dragonian Dual-Reflector." All 17 designs include the Mizugutch feed-axis rotation to reduce cross polarization. You will want to make this one part of your tools for the design of dual reflectors.

The article continues the series by presenting the geometry of the reflector system in all possible combinations of parameters, and gives the equations to calculate every other dimension needed for construction. I wrote these equations into a series of *FORTRAN* subroutines, and I will send them to you by e-mail if you request them by e-mail. I still have the series of subroutines that I wrote for his earlier articles in this column.

Designing Classical Offset Cassegrain or Gregorian Dual-Reflector Antennas from Combinations of Prescribed Geometric Parameters
Part 2: Feed-Horn Blockage Conditions

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1. Introduction

This paper is a short appendix to [1], where classical (unshaped) offset Cassegrain or Gregorian dual-reflector antennas were designed from combinations of prescribed geometric parameters. However, in [1], the model assumed a point-source feed, and did not take into account the physical dimensions of the feed horn needed to illuminate the subreflector. Now, as in [2], we consider a feed horn that can be described in terms of its overall diameter (including the flange) D_f, its length, L_f, and its phase-center position measured from the aperture, D_{pc}. The feed horn is therefore simulated as a cone, which is a fair description in most cases, and is sufficient in the initial design stage of an antenna.

As this is a companion paper, we will refer to [1] instead of repeating here the definitions of the design parameters (see Table 1 of [1]) and the coordinates of the relevant points (see Table 2 of [1]). Figures 1 and 2 of [1] show the geometry of classical Cassegrain and Gregorian dual-reflector antennas, respectively.

2. Feed-Horn Blockage Condition or Feed-Clearance Distance

We will now define the necessary conditions for no feed-horn blockage in terms of geometrical optics, i.e., no ray between the subreflector (SR) and main reflector (MR) is blocked by the horn. When a feed horn, of parameters D_f, L_f, and D_{pc} is used, the geometries of the horn and subreflector are shown in Figure 1 for a Cassegrain system, and in Figure 2 for a Gregorian system. The feed-horn geometry is described by five points (see Figures 1 and 2): F_0, F_a, $F_{a,u}$, $F_{a,l}$, and F_{apex}, which coordinates can be found in Table 1. These points can be useful when drawing the antenna's geometry during the design process.

2.1 Offset Cassegrain System

The feed-horn blockage condition for an offset Cassegrain system (Figure 1) is such that if the vertical distance between the corner, $F_{a,u}$, of the feed horn and the point S, on the lower ray

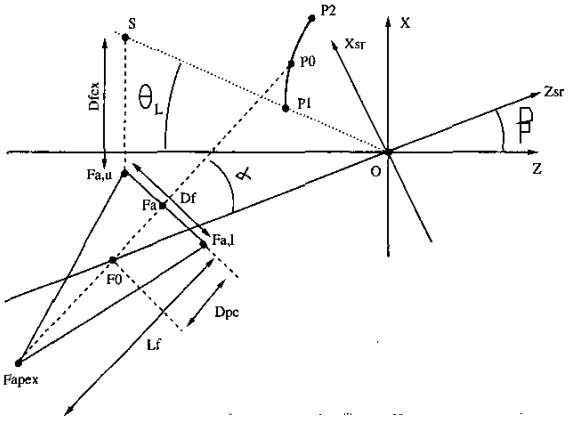


Figure 1. The offset Cassegrain geometry with a conical feed horn.

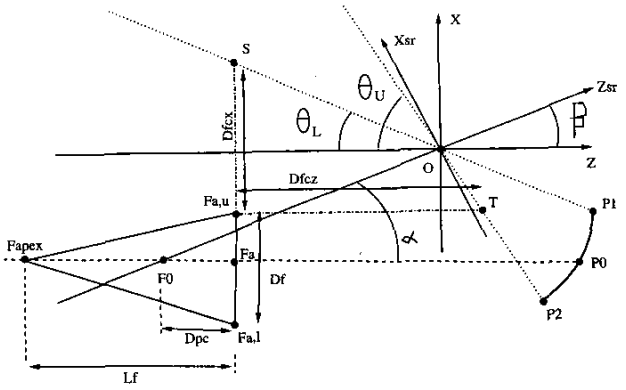


Figure 2. The offset Gregorian geometry with a conical feed horn.

defined by the offset angle θ_L, is positive, then there is no blockage in terms of geometrical optics. The coordinates of S are given in Table 1. The equation for the feed-horn blockage condition is therefore that the feed clearance, D_{fcx}, must be positive:

$$D_{fcx} = \left\{ -2f\cos(\beta) + \sqrt{\frac{D_f^2}{4} + D_{pc}^2} \cos\left[\beta + \alpha + \arctan\left(\frac{D_f}{2D_{pc}}\right)\right] \right\} \tan(\theta_L) + 2f\sin(\beta) - \sqrt{\frac{D_f^2}{4} + D_{pc}^2} \sin\left[\beta + \alpha + \arctan\left(\frac{D_f}{2D_{pc}}\right)\right] \geq 0 .$$

2.2 Offset Gregorian System

There are two feed-horn blockage conditions for an offset Gregorian system. The same condition on D_{fcx} applies, and the same formula as for the offset Cassegrain system can be used to calculate it.

The second feed-horn blockage condition for an offset Gregorian system (Figure 2) is such that if the horizontal distance between the corner $F_{a,u}$ of the feed horn and the point T on the lower ray defined by the offset angle θ_U is positive, then there is no blockage in terms of geometrical optics. The coordinates of T are also given in Table 1. The equation for the second feed-horn blockage condition is therefore that the feed clearance, D_{fcz}, must be positive:

$$D_{fcz} = -2f\cos(\beta) + \sqrt{\frac{D_f^2}{4} + D_{pc}^2} \cos\left[\beta + \alpha + \arctan\left(\frac{D_f}{2D_{pc}}\right)\right] - \|OP_2\|\cos(\theta_U)$$

Table 1. The coordinates of the points.

Point	x	y	z
F_0	$-2f\sin(\beta)$	0	$-2f\cos(\beta)$
F_a	$-2f\sin(\beta)+D_{pc}\sin(\beta+\alpha)$	0	$-2f\cos(\beta)+D_{pc}\cos(\beta+\alpha)$
$F_{a,u}$	$-2f\sin(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\sin\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$	0	$-2f\cos(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\cos\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$
$F_{a,l}$	$-2f\sin(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\sin\left[\beta+\alpha-\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$	0	$-2f\cos(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\cos\left[\beta+\alpha-\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$
F_{apex}	$-2f\sin(\beta)-(L_f-D_{pc})\sin(\beta+\alpha)$	0	$-2f\cos(\beta)-(L_f-D_{pc})\cos(\beta+\alpha)$
S	$-2f\cos(\beta)\tan(\theta_L)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\cos\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]\tan(\theta_L)$	0	$-2f\cos(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\cos\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$
T	$-2f\sin(\beta)$ $+\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\sin\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]$	0	$\|OP_2\|\cos(\theta_U)-\frac{1}{\tan(\theta_U)}\left\{\|OP_2\|\sin(\theta_U)\right.$ $+2f\sin(\beta)$ $\left.-\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\sin\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]\right\}$ with $\|OP_2\|$ given by Equation (34) of [1]

$$+\frac{1}{\tan(\theta_U)}\left\{\|OP_2\|\sin(\theta_U)+2f\sin(\beta)\right.$$
$$\left.-\sqrt{\frac{D_f^2}{4}+D_{pc}^2}\sin\left[\beta+\alpha+\arctan\left(\frac{D_f}{2D_{pc}}\right)\right]\right\}\geq 0$$

with $\|OP_2\|$ given by Equation (34) of [1].

3. Additional Design Options

In [1], 12 design options were presented. Five more design options are presented in Tables 2 and 3, namely options 13 to 17, which contain geometric input parameters related to the profile of the subreflector (a, f, or e) (refer to [1] on how to use these tables). Using these two tables and the equations listed in [1] and in Appendix A of this paper, it is possible to fully describe offset Cassegrain or Gregorian dual-reflector antennas. Design options 13 to 17 allow the antenna designer to specify parts of the subreflector geometry. This can be useful when designing an antenna where the desired input parameters are not listed as design options, i.e., when doing some manual optimizing using the options provided, or for re-designing a main reflector for an existing subreflector.

4. Conclusions

Conditions for the feed-horn blockage in terms of geometrical optics have been derived. Design options additional to those given in [1] for offset dual-reflector Cassegrain or Gregorian antennas were also supplied.

5. Appendix A

Equations (1) to (39) are the same as in [1].

$$\theta_0=\beta-2\arctan\left[\left(\frac{e+1}{e-1}\right)^2\tan\left(\frac{\beta}{2}\right)\right] \tag{40}$$

$$e=\frac{f}{a} \tag{41}$$

$$\theta_U=-2\arctan\left[\tan\left(-\frac{\theta_0}{2}\right)+\frac{Dm}{4F}\right] \tag{42}$$

$$\theta_L=-2\arctan\left[\tan\left(-\frac{\theta_0}{2}\right)-\frac{Dm}{4F}\right] \tag{43}$$

$$A=\tan\left(\frac{\theta_0}{2}\right);$$

$$B=1-\left(\frac{e-1}{e+1}\right)^2;$$

$$C=\left(\frac{e-1}{e+1}\right)^2\tan\left(\frac{\theta_0}{2}\right);$$

Table 2. Design procedure options.

Options	Parameter 1	Parameter 2	Parameter 3	Parameter 4	Parameter 5
No. 13	Dm	F	e	Ls	β
No. 14	Dm	θ_0	θ_e	f	β
No. 15	Dm	F	e	Ds_x	β
No. 16	Dm	F	a	f	β
No. 17	Dm	F	θ_0	a	f

Table 3. Design procedures for options 13 to 17.

Parameter	Option 13	Option 14	Option 15	Option 16	Option 17
Dm	Dm	Dm	Dm	Dm	Dm
F	F	26 (5th)	F	F	F
h	23 (2nd)	23 (6th)	23 (2nd)	23 (3rd)	23 (2nd)
θ_0	40 (1st)	θ_0	40 (1st)	40 (2nd)	θ_0
θ_U	2 (3rd)	3 (4th)	2 (3rd)	2 (4th)	42 (3rd)
θ_L	4 (4th)	4 (7th)	4 (4th)	4 (5th)	43 (4th)
β	β	β	β	β	44 (5th)
θ_e	7 (6th)	θ_e	7 (6th)	7 (7th)	7 (7th)
e	e	5 (1st)	e	41 (1st)	41 (1st)
a	22 (7th)	17 (3rd)	18 (7th)	a	a
f	15 (8th)	f	15 (8th)	f	f
Ds_x	27 (14th)	27 (8th)	Ds_x	27 (8th)	27 (8th)
Ds_y	39 (16th)	39 (16th)	39 (16th)	39 (16th)	39 (16th)
α	6 (5th)	6 (2nd)	6 (5th)	6 (6th)	6 (6th)
Ls	Ls	8 (9th)	8 (9th)	8 (9th)	8 (9th)
Lm	9 (9th)	9 (10th)	9 (10th)	9 (10th)	9 (10th)
d_{sr-mr}	11 (11th)	11 (12th)	11 (12th)	11 (12th)	11 (12th)
d_{f-mr}	10 (10th)	10 (11th)	10 (11th)	10 (11th)	10 (11th)
Lt	12 (12th)	12 (13th)	12 (13th)	12 (13th)	12 (13th)
Ht	13 (13th)	13 (14th)	13 (14th)	13 (14th)	13 (14th)
C_{sr}	38 (15th)	38 (15th)	38 (15th)	38 (15th)	38 (15th)

$$\Delta = B^2 - 4AC$$

$$\beta = 2\arctan\left(\frac{-B \pm \sqrt{\Delta}}{2A}\right) \qquad (44)$$

6. References

1. C. Granet, "Designing Classical Offset Cassegrain or Gregorian Dual-Reflector Antennas from Combinations of Prescribed Geometric Parameters," *IEEE Antennas and Propagation Magazine*, **44**, 3, June 2002, pp. 114-123.

2. C. Granet, "Designing Axially Symmetric Cassegrain or Gregorian Dual-Reflector Antennas from Combinations of Prescribed Geometric Parameters, Part 2: Minimum Blockage Condition While Taking into Account the Phase-Center of the Feed," *IEEE Antennas and Propagation Magazine*, **40**, 3, April 1998, pp. 82-85.

Ideas for Antenna Designer's Notebook

Ideas are needed for future issues of the Antenna Designer's Notebook. Please send your suggestions to Tom Milligan and they will be considered for publication as quickly as possible. Topics can include antenna design tips, equations, nomographs, or shortcuts, as well as ideas to improve or facilitate measurements.